

TEAM - 4

TEARDOWN OF SINAMICS V20

-VARIABLE FREQUENCY DRIVE

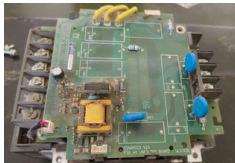
DURGA DEVI K - 23BTT014

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NARESH N - 23BTT310

MUTHU PRASANA - 22BTT017





Product Specifications

Series	Sinamics V20
Power Rating	1.1kw
Input Voltage	440v
Control Method	PWM
Motor Power	1.1kw
Brand	Siemens
Control Mode	V/F Control
Enclosure Type	IP20
Communication Interface	Profibus
Model Name/Number	V20 6SL3210-5BE21-1UV0

INTRODUCTION

- The FSB HV Unfilter Board is connected directly to the AC mains and operates before EMI filtering.
- It processes uncontrolled high-voltage input and generates auxiliary low-voltage supplies required for control electronics.
- The board primarily consists of protection, rectification, switching, and energy storage components.

RESISTORS (R207 MARKINGS)

Small cylindrical axial resistors (with color bands)

Tiny SMD resistors (rectangular, brown/black)

Functions:

- Current limiting
- Protects diodes, ICs, and capacitors from excess current
- Voltage divider
- Scales high voltage down for sensing circuits
- Bleeder resistors
- Discharge capacitors after power-off (important for safety)
- Pull-up / pull-down



CERAMIC CAPACITORS (SMALL BROWN SMD CAPS)

Very small rectangular components near ICs Marked as C2xx, C8xx

Function:

- High-frequency noise suppression
- Decoupling capacitors
- Why needed?
- ICs draw current in fast pulses
- Ceramic caps supply instant current and reduce voltage dips

📌 Without these, the control circuit becomes unstable or noisy.



ELECTROLYTIC CAPACITOR (C811)

One round aluminium capacitor with polarity marking

Function:

- DC smoothing
- Ripple voltage reduction
- Why electrolytic?
- High capacitance in small size
- Used after rectification or DC generation

📌 This capacitor helps maintain steady DC voltage for electronics.



DiODES (V803, V804, V805)

Small black cylindrical / SMD diodes

Functions:

- Rectification
- Converts AC to DC
- Reverse polarity protection
- Flyback protection
- Protects circuits from voltage spikes
- Surge clamping [with MOVs]



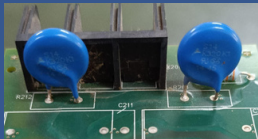
MOVS – METAL OXIDE VARISTORS

Blue Disc Components

Function:

- Absorb sudden voltage spikes
- Protect entire system from:
- Lightning
- Grid disturbances
- Motor switching surges
- Working principle
- Normal voltage → high resistance
- High voltage → suddenly becomes conductive and clamps voltage

📌 Think of MOVs as electrical shock absorbers.



INDUCTOR / CHOKE

Yellow Component

Function:

- Blocks high-frequency noise
- Allows low-frequency power to pass
- Works with capacitors to form LC EMI filter

📌 This prevents the VFD from:

- Polluting the power grid
- Failing EMI/EMC standards



SMALL SMD ICs

Black chips with 5–8 pins

Functions: (based on location & design)

- Voltage regulator
- Converts high DC to low DC (5V / 12V)
- Comparator / monitor IC
- Detects over-voltage / under-voltage
- Protection controller
- Sends fault signals to main controller

📌 These ICs are the “brainlets” of this board.



OPTO-ISOLATION / SIGNAL CONDITIONING (POSSIBLE SMALL ICS)

Why isolation is needed?

This is a high-voltage board and Control electronics must be protected.

Function:

- Transfers signal without direct electrical connection
- Improves safety and noise immunity
- 📌 Even if not obvious, isolation is mandatory in VFD systems.

FERRITE BEADS

Tiny black cylinders or chips

Function:

- Suppress high-frequency EMI
- Act like resistors only for noise

📌 Used on signal lines and power rails.



PCB TRACES & LAYOUT

Observations

Thick copper traces → high current

Wide spacing → high voltage isolation

Clear separation:

HV area and LV control area

- Why layout matters?
 - Prevents electrical breakdown
 - Reduces noise coupling
 - Improves reliability
- 📌 Good layout = half the design success.



CONNECTORS

White JST-type
[japan solderless terminal]

Function Carry:

- Control signals
- Feedback signals
- Low-voltage power



Why connectors instead of soldered wires?

- Easy replacement
- Modular servicing
- Faster assembly

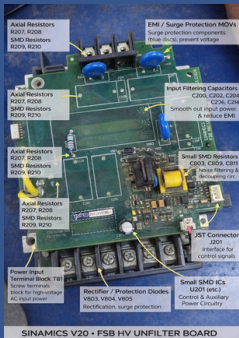


TEST POINTS & MARKINGS

What you see "TESTED" sticker Reference designators.

Purpose:

- Quality control
 - Troubleshooting
 - Automated testing
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-POWER INPUT SECTION

- 1.TERMINAL BLOCK**
- 2.THICK PCB TRACES**

-PROTECTION

COMPONENTS

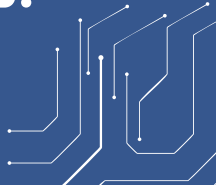
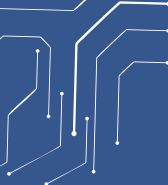
- 1.METAL OXIDE VARISTORS**
- 2.PROTECTION DIODES**

-FILTERING COMPONENTS

- 1.CERAMIC CAPACITORS**
- 2.ELECTROLYTIC CAPACITORS**
- 3.INDUCTORS, CHOKES**
- 4.FERRITE BEADS**

-SIGNAL CONDITIONING COMPONENTS

- 1.RESISTOR**
 - 2.COMBINATION OF (RESISTOR,CAPACITOR,DIODES)**
- SIGNAL CONDITIONING NETWORK**



THANK YOU!